

Product and Category Management System Using Tables

Abstract

This report presents the design and implementation of a **Product and Category Management System** using relational database tables. The system creates **Category** and **Product** tables with primary and foreign key relationships to maintain product information. It supports product insertion, updating, deletion, and category-wise product reporting.

Introduction

A Product and Category Management System is designed to organize products under different categories using database tables. The system ensures efficient storage, retrieval, and management of product data while maintaining data integrity through relationships between tables.

Theory

The system is based on the **Relational Database Management System (RDBMS)** model. Two tables are created:

- **Category Table** – Stores category details.
- **Product Table** – Stores product details and references the Category table through a foreign key.

The relationship between these tables eliminates data redundancy and maintains referential integrity.

Objectives

- Create **Category** and **Product** tables.
- Define **Primary Key** and **Foreign Key** constraints.
- Store product details including name, category, price, and stock.
- Perform INSERT, UPDATE, and DELETE operations on product records.
- Generate category-wise product reports using SQL queries.

Methodology

1. Create the **Category** table with **CategoryID** as the primary key.
2. Create the **Product** table with **ProductID** as the primary key and **CategoryID** as the foreign key.
3. Insert sample category and product records.
4. Update product price or stock when required.
5. Delete unwanted product records.

- Retrieve category-wise product reports using SQL JOIN operations.

Table Structure

Category Table

Column Name	Data Type	Description
CategoryID	INT	Primary Key
CategoryName	VARCHAR(50)	Name of the Category

Product Table

Column Name	Data Type	Description
ProductID	INT	Primary Key
ProductName	VARCHAR(100)	Product Name
CategoryID	INT	Foreign Key referencing Category
Price	DECIMAL(10,2)	Product Price
Stock	INT	Available Stock

Database Operations

The following SQL operations are performed:

- Create Category and Product tables.
- Insert new categories and products.
- Update product information such as price and stock.
- Delete product records.
- Display category-wise product details using SQL queries.

Complexity Analysis

Insertion, updating, and deletion operations execute efficiently using indexed primary keys. Category-wise reports use JOIN operations between the Product and Category tables, providing fast retrieval of related information.

Results

The database successfully stores category and product information, maintains relationships through primary and foreign keys, supports CRUD operations, and generates category-wise product reports accurately.

Conclusion

The Product and Category Management System demonstrates the effective use of relational database tables for inventory management. By organizing data into Category and Product

tables, the system ensures consistency, reduces redundancy, and enables efficient product management and reporting.

References

1. Oracle SQL Documentation.
2. MySQL Reference Manual.
3. Microsoft SQL Server Documentation.
4. Elmasri, R., & Navathe, S. B., *Fundamentals of Database Systems*.